Johnson&Johnson
MedTech

Design Optimization Proposal for Enhanced Robotic Arm Responsiveness

1. Problem Statement

Identified Issues:

1. **Actuator Strain:**
    - **Description:** Actuator strain is when actuators degrade over time during continuous motion due to strain from prolonged usage. Factors that could possibly cause actuator strain are excessive loadings that are handled or lifted by these actuators. Excessive loadings hinder quicker response times because they overcome the actuators normal weight carrying capacity which delays movement.
    - **Impact:** When actuator strain is not handled or even if there is an incident of one, it slows response times. This may have been a contributing factor in the delay of the rotate_joint command that flagged concerns. It delays surgeries and brings much human interaction, the situation that is not wanted. It also reduces efficiency in output.
2. **Sensor Misalignment:**
    - **Description:** Sensor misalignment is a situation that arises from inaccurate positioning of sensors that gather real time feedback. A surgical robotic arm needs constant, accurate, real-time feedback response that ensures that it is well aware of the environment so that it is able to make decisions. However, poorly calibrated and misaligned sensors gather wrong and inaccurate information about positions which hinders accurate signal processing.
    - **Impact:** When the wrong feedback is fed into the processing unit of the surgical robotic arms, it causes inaccurate positioning of the gripper in the surgery process. This allows room for inaccurate positioning of objects and also delays, because it affects response time as well, and the arm becomes unproductive since it does not yield accurate information.

2. Proposed Modifications

Modification 1: Efficiency Improvement

- **Proposal:** The modification involved a function written to improve response time by the use of an efficiency improvement factor. This factor was reduces the response time and hence, allows for faster surgical procedures.
- **Justification:** During my testing, the original response time was 0.2 seconds. In my testing, I used an improvement factor of 0.325. The function run and returned a response time value of 0.135 seconds, decreasing the value by 32.5%. The change is worth considering because it would improve speed of the arm in executing surgical tasks.

Modification 2: Durability Improvement

- **Proposal:** The modification centered on a function that takes in a reinforcement factor and then, uses it to calculate durability performance based on the base durability score. This gives information about how lengthy the surgical robotic arm would perform well.
- **Justification:** During my testing, the original durability was 75 but, with a value of reinforcement factor as 0.625, it skyrocketed the value of durability to 121.88 signalling durability improvement. This score exceeds the 100-point scale of the notebook, so it is illustrative only. The change is necessary to demonstrate how durability might increase if realistic improvements are made.

Additional Recommendation: Sensor Realignment

- Sensor realignment is recommended to address sensor misalignment. It was not simulated, so its benefit is a hypothesis to test.

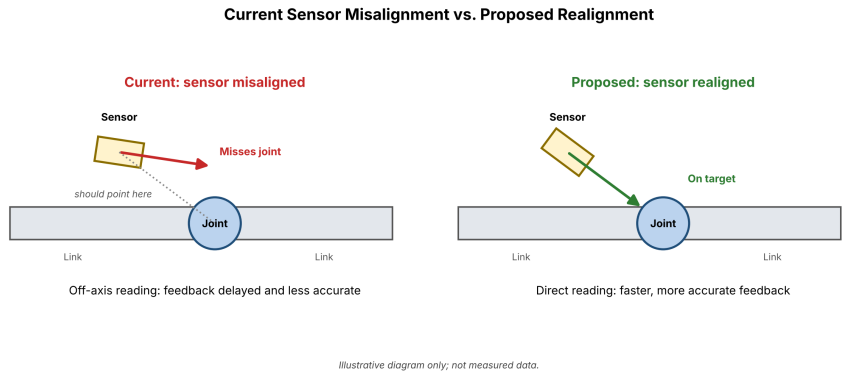
3. Simulation Results

- **Initial Metrics:**
  - **Actuator Response Time:** 0.20 seconds
  - **Sensor Feedback Delay:** Not simulated
- **Post-Optimization Metrics:**
  - **Actuator Response Time:** 0.135 seconds
  - **Sensor Feedback Delay:** Not simulated
- **Key Insights:**
  - The efficiency factor greatly improved response time. It would be very great if we would actualize this improvement by making realistic modification to the actuators and code.
  - The reinforcement factor also demonstrated that, improving durability by probably getting latest hardware or using lightweight but strong materials could boost the surgical arm's performance.

Note: the simulation's overall response time is used as the actuator response time, since the notebook does not separate actuator and sensor delay.

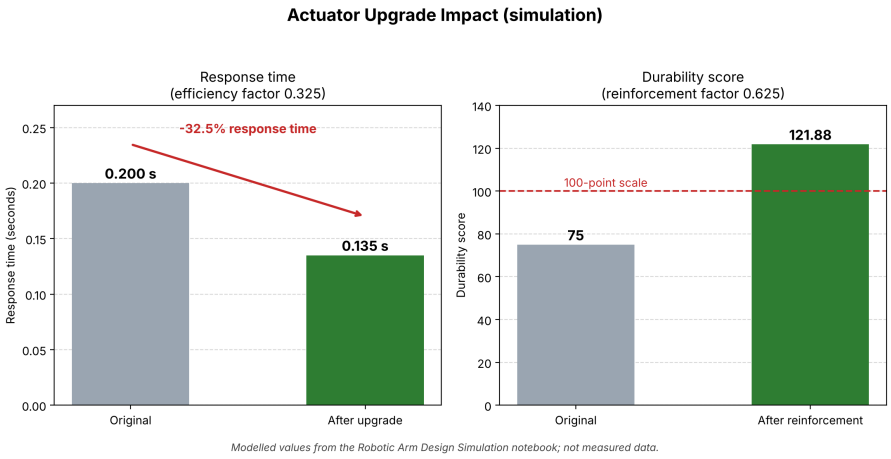
4. Annotated Visual Examples

Visual Example 1: Current Sensor Misalignment vs. Proposed Realignment



Brief Description: An illustrative diagram comparing the current sensor, which points off the joint and gives a delayed, less accurate reading, with the proposed realignment, where the sensor points directly at the joint. The diagram is not measured data.

Visual Example 2: Actuator Upgrade Impact



Brief Description: Bar charts of the simulation results. Response time falls from 0.20 s to 0.135 s (a 32.5% reduction) with an efficiency factor of 0.325, and the durability score rises from 75 to 121.88 with a reinforcement factor of 0.625. The dashed line marks the 100-point scale, which the new durability score exceeds. The values are modelled, not measured.

5. Expected Outcomes

1. **Improved Responsiveness:** Responsiveness is improved by making response time quicker. Responsiveness will also be improved which would curb delays in the surgical procedures. It would also bring surgery accuracy.

2. **Enhanced Reliability:** Enhanced reliability means that there is accuracy in surgery by virtue of the accurate signal information fed into the surgical arm's processing unit. This

ensures that the robotic arm can be trusted as the sensors are accurately aligned for better capture of position.

3. **Increased Durability:** Increased durability talks about improved strength and long lasting efficient performance of the component in the surgical robotic arm. It is brought about from the reinforcement factor which simulates improvement in durability as well as long lasting performance.

6. Conclusion

Summarize the impact of your proposed modifications:

- The proposed efficiency and durability improvements address actuator strain, and sensor realignment is recommended for the sensor issue. In the simulation, an efficiency factor of 0.325 reduced response time from 0.20 s to 0.135 s, a 32.5% reduction, and a reinforcement factor of 0.625 raised the durability score from 75 to 121.88. These values are modelled, not measured. The durability score exceeds the 100-point scale, so the factors need realistic justification before they can be relied on. The effect of sensor realignment was not simulated, so its benefit is a hypothesis to test. If validated, faster and more consistent responses would support more precise positioning during surgery and reduce delays and risk. Next steps are to test realistic factor ranges, check whether added reinforcement slows response, measure sensor feedback delay, and validate both changes on a physical prototype.